GRADE 9 PHYSICS · MOTION

Motion: Distance, Displacement and the Velocity–Time Graph

The same two subtopics, rebuilt for a student who can already solve them.

BAND B80 · high band · bridge retained | GRADE 9 | NO HINTS · POST-HOC SIGNATURES ONLY

WHO THIS BOOK IS FOR

A student who passes probes P1 and P2 without effort and gives a correct prediction on P3, but does not yet reliably name the competing case that would behave differently. That places at 80% — B90 architecture with one adjustment retained: the derivation bridge in M-02 is kept rather than assumed. Nothing here re-explains what distance or a velocity–time graph is. Your remaining errors are method choice under disguise, sign discipline across an axis crossing, and speed.

CONVENTIONS FIXED FOR THIS BOOK

Frame	Motion to the right is positive. The origin is the position at t = 0.
Sign convention	right positive · left negative · time runs forward only
Symbols	d = distance — path length, never negative; s = displacement — vector change in position; v = instantaneous velocity; v_av = average velocity = s / t; t = time
Source ledger	SRC-MOTION-v1
Band placement	P1 ✓ P2 ✓ P3 partial (correct prediction, competing case not named) → 80% → B90 architecture, bridge retained

WHAT IS INSIDE

Subtopic	Title	Concept ID
M-01	Distance and Displacement	PHY-MOT-C01
M-02	Reading a Velocity–Time Graph	PHY-MOT-C03

HOW THIS BAND WORKS

> This book is short on explanation and long on discrimination, because at this level almost every remaining mistake is a correct method applied to the wrong situation. The contrast pairs are the substance: each changes exactly one structural feature and holds everything else fixed, so the boundary becomes visible. The stress set is unlabelled and timed on purpose — retrieval under pressure is the thing being trained.

HOW TO USE IT

Do this	Because
Start at the compression card, not the explanation	There is no explanation here. If you need one, you have been placed in the wrong band.
Write the difference sentence before answering any pair	A right answer for the wrong reason is the failure this book exists to find.
Time the stress set properly	Retrieval under pressure is the skill being trained, not the physics.
Act on the signature, not the score	Same gap twice means serve the repair, not more practice.

M-01 · PHY-MOT-C01

Distance and Displacement

BAND B80 SRC §3.1–3.2

COMPRESSION CARD

The whole subtopic on one card

Cue	The path doubles back, or the question says 'from the start', or a direction is named in the answer
First move	Fix positive direction in words, then track position — not path — leg by leg
Check	$\lvert s\rvert \le d$ always. Equality only when the motion never reverses
Trap	Reporting d when s was asked, whenever any leg runs backwards
Sign	A negative displacement is a direction, not a smaller number

source: SRC-MOTION-v1/§3.1, SRC-MOTION-v1/§3.2

DISCRIMINATION — ONE FEATURE CHANGES

Three pairs. One feature changes in each.

A	B	Differs in	Correct handling
Runner does 4 laps of a 400 m track and stops at the start line.	Runner does 4 laps of a 400 m track and stops half a lap on.	Finishing point only	A: d = 1600 m, s = 0. B: d = 1800 m, s = 2r (track diameter), not 200 m. The path length along the curve is not the displacement.
Walk 30 m east, then 40 m east.	Walk 30 m east, then 40 m north.	Direction of the second leg	A: legs add as numbers, s = 70 m. B: legs add as vectors, s = 50 m at 53° N of E. Same two numbers, different addition rule.
Object returns to its starting point after 12 s.	Object passes its starting point at 12 s and keeps going.	Whether t = 12 s is the end or a moment in passing	A: s = 0 and v_{av} = 0, though the object certainly moved. B: s ≠ 0. Zero average velocity never means zero motion.

For each pair, do not just answer. Write the one sentence that names what changed and why it changes the method. That sentence is the thing being assessed.

source: SRC-MOTION-v1/§3.2

PRACTICE

Recognition drill — 90 seconds, no working

NO HINTS

ID	Question	Type
R1	Runner completes 3 laps of a 250 m track. Displacement?	RECOGNITION
R2	5 m north then 12 m east. Displacement magnitude?	RECOGNITION
R3	An object has non-zero average speed and zero average velocity. Possible or not?	RECOGNITION
R4	$\lvert s\rvert = d$. Under what single condition?	RECOGNITION
R5	Displacement −40 m. State it in words without using the word negative.	RECOGNITION
R6	Distance 0 but displacement non-zero. Possible or not?	RECOGNITION

working space

source: SRC-MOTION-v1/§3.2

BOUNDARY — WHERE STRONG STUDENTS FALL

Where strong students actually lose marks

• Closed loop: d is large, s is exactly zero, and average velocity is zero while average speed is not. Both statements are true at once.

• Displacement quoted without a direction. An unsigned, undirected displacement is an incomplete answer and is marked as such.

• |s| = d is claimed in general. It holds only for motion that never reverses — state the condition when you use it.

• Two-dimensional path treated with one-dimensional arithmetic. Pair 2 above is this failure in its smallest form.

• Negative displacement read as 'less displacement' rather than 'displacement the other way'.

source: SRC-MOTION-v1/§3.2

IF YOU GOT IT WRONG

Error signature routing

What you wrote	What it means	Go to
Correct magnitude, direction omitted	FRAME_SIGN_GAP	PHY-DRILL-SIGN-02
Added perpendicular legs arithmetically	MODEL_SELECTION_GAP	Pair 2, repeated at speed
Said the object was stationary because v_av = 0	CONCEPTUAL_GAP	Boundary item 1
Right on labelled items, wrong in the stress set	TRANSFER_GAP	Extend stress set to 20 items

source: SRC-MOTION-v1/§3.2

RELEASE CONDITION FOR M-01
Clean across the recognition drill and all three pairs, with the difference sentence written correctly each time, releases you to M-02. A correct answer with a wrong or missing difference sentence does not — that is the pattern that plateaus strong students, because the score looks fine while the discrimination is not there.

M-02 · PHY-MOT-C03

Reading a Velocity–Time Graph

BAND B80 SRC §3.4

COMPRESSION CARD

The whole subtopic on one card

Cue	Velocity is on the vertical axis and the question asks how far
First move	Split the region under the line into rectangles and triangles, then sum with sign
Check	Units resolve to metres; the sign agrees with the described direction of travel
Trap	Reading height where area is meant — or returning distance when displacement was asked, whenever any part of the line drops below the axis
Slope vs area	Slope of v–t is acceleration. Area of v–t is displacement. The axis label decides which question you are being asked

source: SRC-MOTION-v1/§3.4

Bridge retained. At 80% this derivation is usually held rather than owned, so it is kept rather than assumed. For a straight ramp from rest to v over time t, the average velocity is v/2 because the velocity rises evenly, so half of it is as good as all of it for the purpose of covering ground. Then $s = v_{av} \times t = \frac{1}{2}vt$, which is precisely the triangle's area. The triangle formula is not a separate rule to remember — it is the rectangle rule with the average substituted for the constant.

source: SRC-MOTION-v1/§3.4

PLACEHOLDER FIG-M02-B80-01 status=RESERVED

- Two panels side by side, identical outline shape in both — this is the entire point
- Panel A: axes labelled 'velocity (m/s)' and 'time (s)'; shape rises 0→8 over 4 s, holds 8 to 7 s
- Panel B: axes labelled 'distance (m)' and 'time (s)'; shape geometrically identical to A
- Both panels annotated at t = 5 s with a small dot on the line, no value given
- No shading, no area highlighting — the reader must decide what to do, not be shown
- Caption must not reveal which reading applies to which panel

reserve 56mm · src SRC-MOTION-v1/§3.4

Same shape, twice. At t = 5 s, describe the motion in each panel.

source: SRC-MOTION-v1/§3.4

DISCRIMINATION — ONE FEATURE CHANGES

Three pairs. One feature changes in each.

A	B	Differs in	Correct handling
FIG panel A — vertical axis reads velocity.	FIG panel B — vertical axis reads distance, shape identical.	Axis label only	A at t = 5 s: object moving at a steady 8 m/s, zero acceleration, and distance is accumulating as area. B at t = 5 s: object at rest, because the line is flat and slope is the speed. Identical picture, opposite physical story.

A	B	Differs in	Correct handling
v–t line above the axis throughout, asked for distance.	Same shape with the second segment below the axis, asked for displacement.	Sign of one segment	A: sum the areas. B: subtract the area below the axis. Ask for distance in B instead and you add magnitudes — one word in the question changes the arithmetic.
Straight ramp from rest to 12 m/s over 4 s.	Curve from rest reaching 12 m/s at the same 4 s, concave down.	Straight versus curved path between the same endpoints	Same endpoints, different areas. The concave curve lies above the chord throughout, so it has covered more ground. State which is larger without computing — that is the skill being tested.

Name the changed feature in one sentence per pair before giving any answer. A correct answer with the wrong reason counts as wrong here.

source: SRC-MOTION-v1/§3.4

STRESS SET — UNLABELLED, TIMED

Eight items · unlabelled · mixed across both subtopics

12 minutes. No method names, no headings, no working shown to you. Mixed deliberately — deciding which idea applies is the assessment.

#	Question
1	A body moves at 6 m/s for 4 s, then reverses at 3 m/s for 4 s. Give both the distance and the displacement.
2	A v–t graph is a horizontal line at −5 m/s from 0 to 6 s. Describe the motion and find the displacement.
3	The area under a v–t graph between 0 and 10 s is zero. State two different things the object may have done.
4	A d–t graph is horizontal between 3 s and 7 s. What is the object doing?
5	A car ramps from rest to 20 m/s in 5 s, then holds it for 5 s. Total distance?
6	Two v–t graphs have equal areas but different shapes. What quantity must be equal, and what need not be?
7	An athlete runs one full lap of a 400 m track in 50 s. Give the average speed and the average velocity.
8	A v–t line crosses the time axis at t = 4 s. What is happening at that instant, and what happens to the displacement after it?

Mark into four buckets: correct · correct-by-luck · wrong-method · execution-slip. The middle two generate the signatures worth acting on; two buckets would hide them.

source: SRC-MOTION-v1/§3.2, SRC-MOTION-v1/§3.4

BOUNDARY — WHERE STRONG STUDENTS FALL

Where strong students actually lose marks

• Velocity crosses zero mid-graph: displacement subtracts, distance adds. The question's single word decides it.

• Area sums to zero and the object has certainly moved — zero displacement is not zero motion.

• A vertical jump in a v–t line would mean infinite acceleration, so no real graph has one. Say why when you meet one.

• Velocity negative throughout: the area is negative and the sign carries the direction, not an error.

• Reading a value off the line when the question asked about the region underneath it, or the reverse.

source: SRC-MOTION-v1/§3.4

WHERE THIS GOES NEXT — NOT ASSESSED

You have been finding an area in order to get a distance. The same move works far more generally than you have been told: the area under a graph of **any rate against time** gives the total accumulated amount. Area under a flow-rate graph gives the volume that flowed. Area under an acceleration–time graph gives the change in velocity.

Everything you have used so far assumed the acceleration stays constant. When it does not, the equations you know stop working — but the area does not. That is why the graph method outlives them.

Try it: A body accelerates steadily at 3 m/s² for 4 s. Sketch its acceleration–time graph, find the area, and say in words what that number is. (It is 12 m/s — the change in velocity. You have just done something you will be formally taught in Grade 11.)

IF YOU GOT IT WRONG

Error signature routing

What you wrote	What it means	Go to
Correct magnitude, sign wrong after the axis crossing	FRAME_SIGN_GAP	PHY-DRILL-SIGN-02
Used slope where area was required	MODEL_SELECTION_GAP	Pair 1, repeated at speed
Gave distance when displacement was asked	MODEL_SELECTION_GAP	Pair 2 + boundary item 1
Correct on labelled work, wrong in the stress set	TRANSFER_GAP	Extend stress set to 20 items, mix in 3 subtopics
Triangle area arithmetic slipped	EXECUTION_GAP	PHY-DRILL-AREA-01 — do not re-teach the concept

source: SRC-MOTION-v1/§3.4

SELECTION RULE FOR WHOEVER IS MARKING THIS

Same gap twice on one concept → serve the repair object, not more practice. Same gap across three different concepts → it is a skill deficit, not a concept deficit; serve the cross-cutting drill instead. Clean across the unlabelled stress set → release to the next concept or to the preview. More practice on a gap that has already appeared twice is the most common way strong students plateau.

APPENDIX

Source, link and badge ledger

Every teaching block traces to a source obligation or is declared VALUE_ADD. Nothing in this book is unattributed.

Subtopic	Block	Source obligation
M-01	compression_card	SRC-MOTION-v1/§3.1, SRC-MOTION-v1/§3.2
M-01	discrimination	SRC-MOTION-v1/§3.2
M-01	practice	SRC-MOTION-v1/§3.2
M-01	boundary_set	SRC-MOTION-v1/§3.2
M-01	gap_map	SRC-MOTION-v1/§3.2
M-01	callout	VALUE_ADD
M-02	compression_card	SRC-MOTION-v1/§3.4
M-02	prose	SRC-MOTION-v1/§3.4
M-02	figure	SRC-MOTION-v1/§3.4
M-02	discrimination	SRC-MOTION-v1/§3.4
M-02	stress_set	SRC-MOTION-v1/§3.2, SRC-MOTION-v1/§3.4
M-02	boundary_set	SRC-MOTION-v1/§3.4
M-02	preview	VALUE_ADD
M-02	gap_map	SRC-MOTION-v1/§3.4
M-02	callout	VALUE_ADD

Figure placeholders awaiting artwork

Figure	Status	Reserved	Caption
FIG-M02-B80-01	RESERVED	56 mm	Same shape, twice. At t = 5 s, describe the motion in each panel.

Audit checkpoints cleared at render time

Checkpoint	Result
BAND DECLARED (SRU-23)	PASS — B80 high band, placement recorded on cover
BAND FIDELITY D1	PASS — no re-teaching block present; opens at compression in both subtopics
FRAME/SIGN BEFORE ALGEBRA (SRU-24)	PASS — declared on cover, held throughout
SYMBOL DRIFT D2	PASS — v_av introduced in frozen table, used consistently
VOCABULARY DRIFT D8	PASS — distance/displacement, speed/velocity strict throughout
HINT POLICY	PASS — zero hints; feedback is post-hoc signature routing only
CONTRAST PAIR CLOSENESS	PASS — all 6 pairs change exactly one structural feature
STRESS SET UNLABELLED	PASS — 8 items, no headings, no method names, timed
PREVIEW FENCED (§4.1)	PASS — scored:false, formalism unnamed, excluded from mixed mastery and coverage denominators
MODEL UPGRADE RECORDED (SRU-25)	PASS — constant-acceleration limit stated in preview
SOURCE LINKAGE	PASS — 11 of 14 blocks cite an obligation; 3 declared VALUE_ADD
FIGURE PLACEHOLDERS	1 reserved, bounds fixed, semantic spec forbids revealing the answer in the caption
BAND PRACTICE MIX	PASS — 20% recognition speed drill present per B90 contract